Solution of Triangle

Mind map

Basics

- $A + B + C = \pi$
- $|b - c| < a < b + c$

Sine rule

- $\frac{a}{\sin A} = 2R$
- Identities: $a = k \sin A$
- Split by a cevian

Cosine rule

- $\cos A = \frac{b^2+c^2-a^2}{2bc}$
- Nature: $a^2 + b^2 \gtrless c^2$
- $\cot A = \frac{b^2+c^2-a^2}{4\Delta}$

Projection, tangent rule

- $a = b \cos C + c \cos B$
- $\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$

Area

- $\frac{1}{2}bc \sin A = \frac{abc}{4R} = rs$
- Heron
- Altitude $\frac{2\Delta}{a}$

Half angles

- $\tan \frac{A}{2} = \frac{\Delta}{s(s-a)}$
- $\cos^2 \frac{A}{2} = \frac{s(s-a)}{bc}$

Ambiguous case

- b, c, B given
- 0 / 1 / 2 / 1 triangles

m-n theorem

- $(m + n) \cot \theta = m \cot \alpha - n \cot \beta$
- $= n \cot B - m \cot C$

Circumcircle

- $R = \frac{abc}{4\Delta}$
- O to BC: $R \cos A$
- $\angle BOC = 2A$

Incircle

- $r = \frac{\Delta}{s}$
- Tangents $s - a$
- $\angle BIC = \frac{\pi}{2} + \frac{A}{2}$

Excircles

- $r_1 = \frac{\Delta}{s-a} = s \tan \frac{A}{2}$
- $\sum r_1 = 4R + r$
- $\sum r_1 r_2 = s^2$

Median, bisector

- $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$
- $AD = \frac{2bc}{b+c} \cos \frac{A}{2}$
- $BD : DC = c : b$

Orthocentre, pedal

- $HA = 2R \cos A$
- Pedal sides $a \cos A$
- Pedal $R' = \frac{R}{2}$

Excentral triangle

- Angles $\frac{\pi}{2} - \frac{A}{2}$
- Sides $4R \cos \frac{A}{2}$

Regular polygon

- $a = 2R \sin \frac{\pi}{n} = 2r \tan \frac{\pi}{n}$
- Area $nr^2 \tan \frac{\pi}{n}$

Special figures

- Equilateral: $R = 2r$
- Cyclic quad: $\sqrt{\prod(s - a)}$